Week 6 Report: SAR-Only Crop Classification (Group 11)

1. Label definition and class counts

Country	Corn	Soybean	Wheat	Total
Argentina	134	409	122	665
Brazil	116	289	136	541
Both	250	698	258	1,206

2. Final field list and dropped fields

The final list has 1,206 fields (665 Argentina, 541 Brazil), chosen from the 1,302 Argentina and Brazil fields in the raw JSON inventory. A field was retained when the field was downloaded, co-registered and successful for all the automatic checks. CRS match, raw centroid in field, percentage of valid pixels ≥ 90% and plausible VV and VH.

Country	Raw fields	Kept	Failed checks	No usable SAR
Argentina	751	665	12	74
Brazil	551	541	10	0
Total	1,302	1,206	22	74

The 74 fields in Argentina that do not have useful SAR data are 67 that do not have any SAR acquisition during the sowing to harvest stage, and 7 that had one acquisition but was unable to be co-registered. There were only less than 90% valid pixels on all 22 failed fields.

3. Spatial fold design

The data in 1206 fields of data is divided into 5 folds, meaning no 10 km block in the data is used for both training and test data. The field centroids were transformed to kilometres per country and clustered in complete-linkage using a 10 km threshold, resulting in 76 blocks. Then, whole blocks were allocated with StratifiedGroupKFold (5 folds stratified on country/crop, shuffled, seed 42). The typical program of leakage checking identified 0 blocks that are inside both training and testing. Whole blocks move together and make folds of 173 to 322 fields. This is important because neighbouring fields have similar soil conditions, weather, management and may have the same radar scene. A random partition would allow the model to "learn" a test field from its "close" training neighbours and overestimate the accuracy.

4. SAR preprocessing

This is provided as one multi-date GeoTIFF per field for Sentinel-1 IW, VV and VH in dB, 10 m pixels, one relative orbit and co-registered through phase cross-correlation to the median image of the field (UTM zones 20S to 22S).

Fixed 11x11 (110 m) square Field Mask centred at the raw JSON centroid. The yield-masks are not included in the downloaded tiles, so the planned fallback was applied to each field. Averaging: Pixel values are first converted from dB to linear power (10dB), and then the average of all pixels for the window is calculated, then converted back to dB. The mean would be averaged and would be biased downward in dB. Week of the season: (date − sowing) / (harvest − sowing). The dates of tile acquisitions are not stored in the band names, so for 1205 fields, the dates of the first and last Sentinel-1 date in the download log are used, and for 1 field, the dates of sowing and harvesting. The dates given are thus the nearest available approximations.

Season bins: 6 equal size bins of season fraction The average dates of each bin are calculated in linear power. Linear interpolation to fill empty bins.

Output: sar_timeseries.csv - one row per field, bin: VV and VH mean and standard deviation, and flag for interpolated bins.

5. SAR feature definitions

The 42 features are saved to a file called sar_features.csv, with each field having a prefix sar_ and being identified by a field_id.

Five per-date series are built from the field's VV and VH values: VV (dB), VH (dB), VH−VV (dB), VH/VV (linear ratio) and RVI = 4·VH / (VV + VH) with VH and VV in linear power. The season summary includes for each series the mean, standard deviation, maximum, minimum, amplitude (max – min) and position of the peak in the season (0-1). That gives 5 × 6 = 30 features.

The other 12 are the 6-bin season profile of VV and 6-bin season profile of VH (sar_vv_bin1 to 6, sar_vh_bin1 to 6).

The canopy structure and volume scattering cause the responses of VH−VV and RVI, and soil moisture causes the response of VV. GLCM texture is available, but not calculated. The best model that only uses SAR is RBF-kernel SVM with an accuracy of 68.5% and macro-F1 0.867 on 1,206 fields.

6. SAR-only results

The best model using only SAR data is RBF-kernel SVM with 1,206 fields, accuracy of 88.5% and macro-F1 0.867. All scores are 5-fold Cross Validation, missing values were imputed by the median and the class weights balanced.

Model	Accuracy	Macro-F1	Weighted-F1
Logistic Regression	0.866	0.847	0.867

Model	Accuracy	Macro-F1	Weighted-F1
Random Forest	0.886	0.861	0.885
SVM (RBF)	0.885	0.867	0.886
XGBoost	not run	not run	not run

XGBoost was ignored due to being too large to be installed online. All three models are within 0.02 macro-F1 of each other, and they were executed only once, therefore, no one model is significantly better.

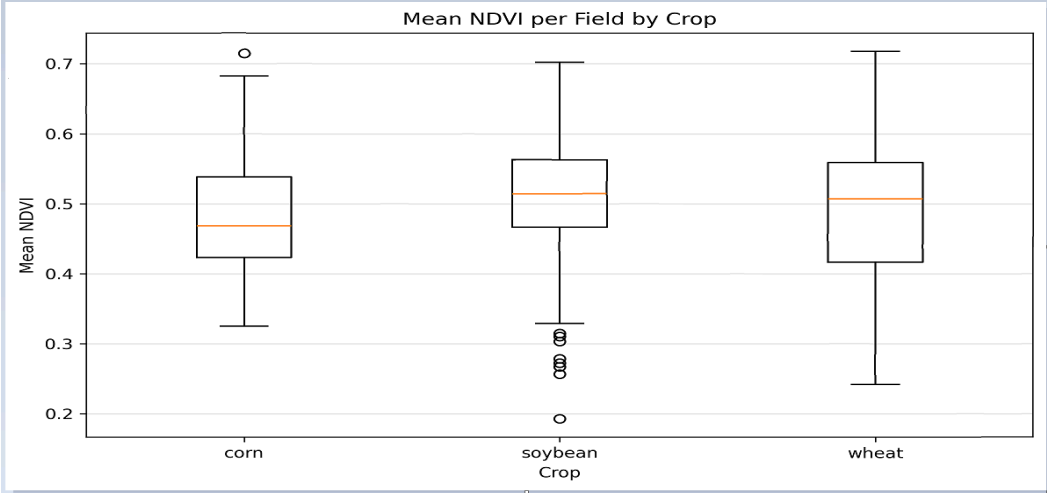
Confusion matrix of the best model (rows are true crops, columns are predicted):

True \ Predicted	Corn	Soybean	Wheat
Corn	199	36	15
Soybean	57	625	16
Wheat	10	5	243

Wheat is recognised best (243 of 258, 94%), then soybean (625 of 698, 90%) and corn (199 of 250, 80%). The most common mistakes (93 fields) are for corn and soybeans to be misidentified as the other crop. They are both summer crops with a very similar canopy and so the radar backscatter is similar; wheat crops have a different season and structure. This is where optical data like NDVI is supposed to correct the imbalance in the fusion step. In order to analyse the optical characteristics, the following procedure was conducted.

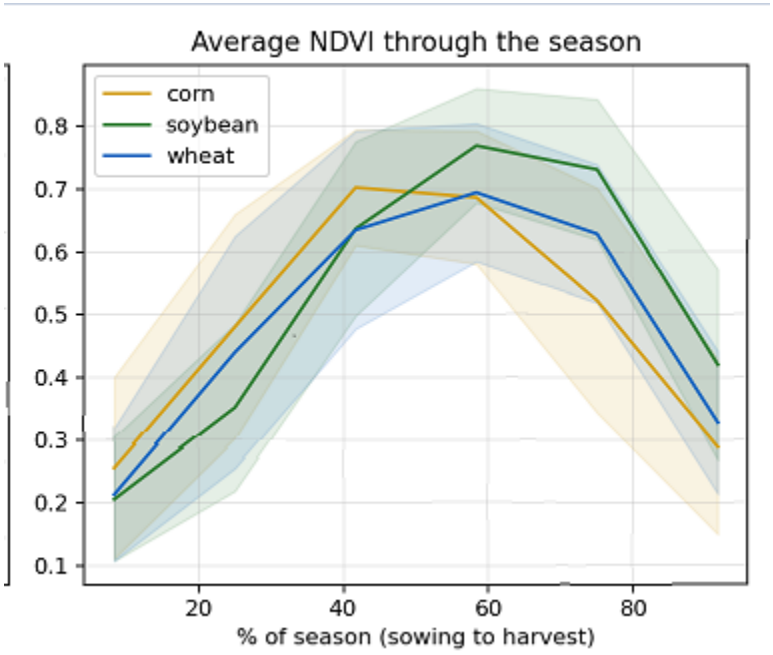
7. Sentinel-2 Optical Feature Results and Analysis

Mean NDVI values of 1206 agricultural fields belonging to three crop types (corn, soybean and wheat) were calculated. We made a box plots based on the latest Sentinel-2 feature dataset by the NDVI values in order of the crop.



The above image shows mean NDVI across different croptypes.

The average NDVI was 0.483 for corn is 0.515 for soybean and about 0.490 for wheat. Soybean had the maximum mean NDVI value while wheat and corn had least values. However, there are overlaps in the distributions, and NDVI may not be adequate to discriminate among all three crop types. The early observations aid me in noting some differences in the characteristics of the vegetation as seen under corn, soybean and wheat. They give a foundation for later combining optical features with SAR features, in the fusion stage.



The crop-wise optical results showed the following patterns:

Crop	Mean NDVI	Mean peak NDVI	Mean NDRE	Mean NDWI
Corn	0.484	0.779	0.398	-0.446
Soybean	0.515	0.822	0.405	-0.483
Wheat	0.491	0.756	0.391	-0.452

8. Optical-only Classification Results

Three machine-learning models (Logistic Regression, Random Forest and Support Vector Machine (SVM)) were tested with the 36 optical features derived from Sentinel-2. Same Spatial 5-fold cross-validation methodology was used to evaluate the models.

Model	Accuracy	Macro-F1	Weighted-F1
Logistic Regression	0.884	0.862	0.885
Random Forest	0.780	0.728	0.780
SVM	0.847	0.815	0.847

The highest accuracy and Macro-F1 score of 88.4% and 0.862, respectively, were obtained using Logistic Regression with optical data. SVM also performed satisfactorily and Random Forest performed comparatively low.

The results showed that the vegetation features from Sentinel-2 can be used effectively to differentiate corn, soybean and wheat. There is still some confusion within the different crop classes, however.

9. SAR–Optical Feature Fusion Results

To explore the effectiveness of the integrated use of Sentinel-1 and Sentinel-2 features for improved crop classification.

The five-fold spatial cross-validation method was used for all three approaches, applying 1206 fields. All fused models outperformed the other approaches in terms of Macro-F1 score. The highest accuracy and Macro-F1 score for the fused features was obtained by Logistic Regression at 95.8% and 0.948, respectively.

The Random Forest feature importance analysis showed that around 63% of the total feature importance is attributed to the SAR features and 37% of the total feature importance is attributed to optical features.

The results have indicated that a fusion of radar and optical data can yield complementary information on crop parameters which can increase classification accuracy beyond that achieved using either data set alone.

Model	SAR-only Macro-F1	Optical-only Macro-F1	Fused Macro-F1
Logistic Regression	0.847	0.862	0.948
Random Forest	0.861	0.728	0.895
SVM	0.867	0.815	0.914

10. Discussion of Optical and Fused Results

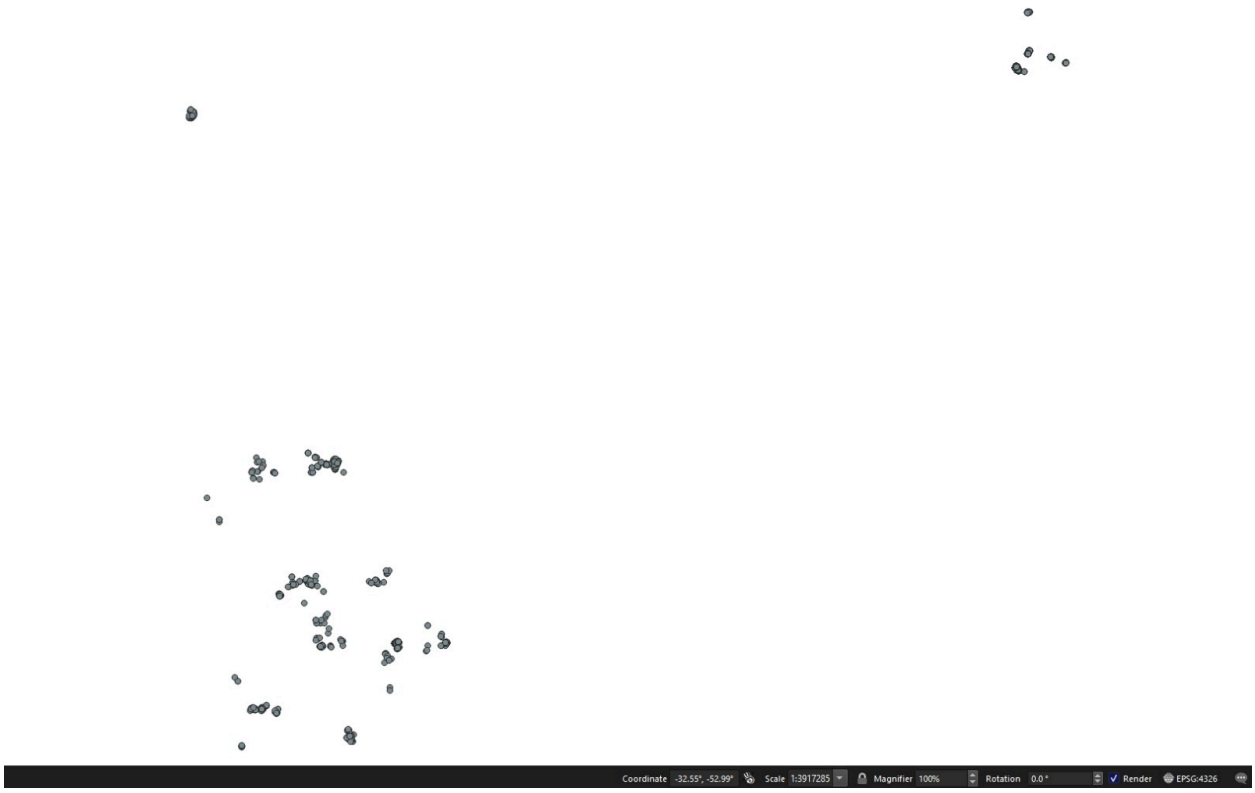
From the optical analysis, the vegetation indices obtained from image data of sentinel-2 delivered useful information on the crop growth and seasonal changes, with logistic regression, the best optical only accuracy was obtained (88.4%). The combination of Sentinel-1 and Sentinel-2, has enhanced classification accuracy, with Logistic Regression attaining 95.8% accuracy and a Macro-F1 score of 0.948. This suggests that the spectral information from the optical sensors can be used with the SAR data, so that general classification performance can be enhanced. The optical analysis revealed that Sentinel-2 vegetation indices can be used as useful data for the crop growth and seasonal changes and Logistic Regression gave the highest optical-only accuracy of 88.4%. The combination of Sentinel-1 and Sentinel-2 provides additional benefits in terms of classifiability, such as Logistic Regression with an accuracy of 95.8% and Macro-F1 of 0.948.

11. Resolved Week 5 limitations

- **Seven fields not co-registered.** 7 fields not co-registered. There are no "second images" to align, that is, all are Argentina 2022 fields with just one Sentinel-1 acquisition taken somewhere between sowing and harvest time.

- **7.4 m maximum offset.** The offset between the raw JSON centroid and the tile centre is a maximum of 7.4 m, mean of 2.9 m, across all 1,206 fields, and within a 10 m pixel, within the TA's 2-4 pixel tolerance. Pixel snapping and not misalignment, as the "centroid" may not change more than ±½ pixel diagonal (approx ±7.1 m) from the snapping to the 10 m pixel grid.
- **22 failed fields** (12 Argentina, 10 Brazil).All had under 90% valid pixels (minimum 0.52, median 0.78). They are flagged and dropped, not downloaded, of the fields are 1.8%.
- **67 fields with no data.** No VV+VH acquisition in their window: All Argentina 2022 (37) and 2023 (30) fields. All 74 lost Argentina fields are from 2022 and 2023, which uses 7 single-date fields. We have not confirmed the loss of Sentinel-1B in December 2021 or the low number of dual polarization acquisitions in Argentina. As a restriction, however, the most recent seasons are not well represented in Argentina.
- **Co-registration quality.**20,235 image dates, 50 of which were over 50 m, were measured, of which 20,235 were over 20 m.A total of 20,235 image dates were measured and 50 of them were above 50 m, which is implausible for Sentinel-1 and suggests miscorrelation of low-contrast tiles. No shifting occurred of these dates. The median of the applied shifts is 3.6 m. The features average 110 m window size, which has a small effect on the results, as the dates do not agree with each other.
- **QGIS check.** he raw JSON centroids and the SAR tile footprints were overlaid in QGIS. All the centroids are shown to be within their tiles (see attached screenshots).





- **Materials and Tools.** Python 3.11, pandas, NumPy, SciPy, rasterio, scikit-image and scikit-learn; Google Earth Engine to retrieve Sentinel-1 and Sentinel-2; QGIS for an overlay check. Optical bands: Sentinel-2 B4 (red) and B8 (near-infrared) for NDVI, B5 for NDRE and B3 for NDVI.

12. Optical progress and plan for Weeks 7-8

Optical status: optical preparation (Task 4) is included, including a Sentinel-2 cloud mask, NDVI, NDRE and NDWI per field, the same field mask and season bins as SAR, a box plot of NDVI per class. In the previous results reported from this mid-semester presentation, only 100 preprocessed records were used (NDVI mean 0.357 and NDVI standard deviation 0.148) and their IDs did not match the raw field IDs and hence are not used for fusion.

Plan for Weeks 7 and 8:

1. Complete s2_timeseries.csv and the corresponding s2_ feature table, which have all 1,206 fields keyed by field_id.
2. Use the same folds.csv to train optical-only models, using the same evaluation script.
3. Fuse is to join the Sar_ and S2_ feature tables on the field_id and training together.
4. Discuss how accurately they classify the crop, macro-F1, macro-weighted-F1, and a per-crop score and compare this with a paired test comparing the same models across folds, and determine if the fused models reduce the corn vs soybean errors.